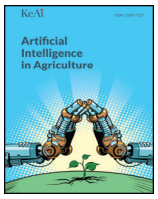
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Boosting grapevine phenological stages prediction based on climatic data by pseudo-labeling approach

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ABSTRACT

Predicting grapevine phenological stages (GPHS) is critical for precisely managing vineyard operations, including plant disease treatments, pruning, and harvest. Solutions commonly used to address viticulture challenges rely on image processing techniques, which have achieved significant results. However, they require the installation of dedicated hardware in the vineyard, making it invasive and difficult to maintain. Moreover, accurate prediction is influenced by the interplay of climatic factors, especially temperature, and the impact of global warming, which are difficult to model using images. Another problem frequently found in GPHS prediction is the persistent issue of missing values in viticultural datasets, particularly in phenological stages. This paper proposes a semi-supervised approach that begins with a small set of labeled phenological stage examples and automatically generates new annotations for large volumes of unlabeled climatic data. This approach aims to address key challenges in phenological analysis. This novel climatic data-based approach offers advantages over common image processing methods, as it is non-intrusive, cost-effective, and adaptable for vineyards of various sizes and technological levels. To ensure the robustness of the proposed Pseudo-labelling strategy, we integrated it into eight machine-learning algorithms. We evaluated its performance across seven diverse datasets, each exhibiting varying percentages of missing values. Performance metrics, including the coefficient of determination (R^2) and root-mean-square error (RMSE), are employed to assess the effectiveness of the models. The study demonstrates that integrating the proposed Pseudo-labeling strategy with supervised learning approaches significantly improves predictive accuracy. Moreover, the study shows that the proposed methodology can also be integrated with explainable artificial intelligence techniques to determine the importance of the input features. In particular, the investigation highlights that growing degree days are crucial for improved GPHS prediction.
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1. Introduction

Grapevine phenological stages (GPHS) are important indicators of agricultural operations such as pruning, pest control, soil tillage, and harvest. A profound comprehension of the impact of climatic factors, with a particular emphasis on temperature, is essential for the effective management of viticulture (Estrella et al., 2007). Nevertheless, the intricate interplay with variables such as cultivars, rootstock, soil type, and atmospheric conditions adds complexity to the prediction of grapevine phenological development (Sabbatini and Howell, 2013). Global warming also aggravates this complexity, leading to shifts in phenological stages across various crops, including grapevines. For instance, in Spain, it was estimated that by 2050, bud-break and flowering for cultivars such as Verdejo, Viura, and Sauvignon Blanc could occur seven days earlier (Ramos and Yuste, 2023), highlighting the impacts of climate change on viticultural timelines.

The anticipated acceleration of the spring season by 2–5 days per decade (Tomasi et al., 2011) presents a significant challenge for grapevine cultivation, with potential repercussions for berry composition and, consequently, wine quality (Mira de Orduña, 2010; Ovadia et al., 2013). Recent studies highlight that such phenological shifts threaten current grapevine cultivation areas, underscoring the need for proactive measures (e.g., Sodini et al., 2023). Observations in Conegliano, Italy, from 1964 to 2009 revealed a shift of up to 19 days in phenological stages, with an average increase of 8 days per 1.0 °C rise in temperature for bud-break, bloom, veraison, and maturation (Tomasi et al., 2011). While cultivation techniques and varietal resilience have demonstrated some capacity to mitigate the impacts of global warming, viticultural regions like Bordeaux and Napa have experienced phenological shifts over the last 60 years that are no longer favorable to producing high-quality wines (Kurtural and Gambetta, 2021).

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Machine learning (ML) has become essential for deciphering intricate dynamics and addressing the aforementioned challenges, particularly within agricultural domains (Cravero et al., 2022). ML and control systems have transformed precision agriculture by enabling data-driven insights that optimize resource use, crop management, and sustainability, utilizing environmental sensors, satellite imagery, and climate data for real-time predictions and control (Kocian and Incrocci, 2020). Recent advancements in deep learning (DL) have further empowered precision farming with neural networks that process images and time-series data, enabling applications like convolutional neural networks (CNNs) for crop health monitoring and recurrent neural networks (RNNs) for temporal predictions (Coulibaly et al., 2022). This also led to improvements in phenological stage prediction for crops like rice, trees, and olives. For instance, Qin et al. (Qin et al., 2023) used deep learning with ResNet-50 for rice, achieving high accuracy in stage recognition from imagery. In tree phenology, Sainte et al. (Sainte et al., 2024) introduced the PhenoFormer model, which adapts effectively to climate variability and outperforms traditional models in prediction accuracy. In viticulture, DL techniques have been developed for grape bunch detection (e.g., Aguiar et al., 2021; Schieck et al., 2023) and variety recognition (Franczyk et al., 2020). The primary focus of grape stage prediction research is identifying growth and maturation phases through image processing, primarily by distinguishing grape colors (Schieck et al., 2023). However, these methods depend on image processing and require cameras or sensors in vineyards, which can be costly and inconvenient to maintain, especially in large areas (Seng et al., 2018; Tardaguila et al., 2021).

In addition to the costly maintenance challenge, data collection is another critical issue encountered in addressing the GPHS problem. In viticulture, data collection is challenging due to the need for fieldwork to check the stage, resulting in a small amount of annotation during the viticultural timeline and missing values on most days within the grapevine phase, significantly degrading the quality of the machine-learning solution's estimations (Palanivinayagam and Damaševičius, 2023). To address missing data, which is commonly encountered across various domains, including natural language processing (Yang et al., 2023), computer vision (Ciarfuglia et al., 2022), and speech recognition (Zhang et al., 2022), semi-supervised learning methods have emerged as effective solutions.

Semi-supervised learning methods become particularly valuable in scenarios where the availability of labeled data is limited, with this constraint posing challenges for conventional supervised learning methods (van Engelen and Hoos, 2020). While few studies applied semi-supervised learning approaches in agriculture-related domains (e.g., precision agriculture (Ciarfuglia et al., 2022), fruit detection and segmentation (Ciarfuglia et al., 2023), semantic segmentation in vineyards (Casado-García et al., 2022)), none have addressed the problem of missing values in GPHS annotations.

To tackle these two challenges, our proposed approach in this paper, called Pseudo GPHS Labelling, relies on readily available climatic data, eliminating the need for additional hardware and reducing the associated expenses while also introducing pseudo-labeling for GPHS to address the missing values in the target data. By doing so, our approach is scalable and adaptable for vineyards of various sizes, locations, and even less technologically advanced regions, unlike image-based systems that may struggle with vast vineyard areas (Seng et al., 2018). Our approach can also be integrated with standard interpretability tools, such as SHapley Additive exPlanations (SHAP), enabling us to explore the effect of the analyzed characteristics on the final prediction and identify those having the most significant impact. The insights derived from these tools can prove valuable for both researchers and domain experts, highlighting patterns in data and assisting them in making informed decisions or developing further hypotheses for investigation. To demonstrate the effectiveness of the proposed solution, we selected a set of eight machine learning algorithms and performed a comprehensive evaluation using two key error metrics: coefficient of determination (R^2) and root-mean-square error (RMSE). To ensure a robust assessment, we conducted this evaluation across seven distinct datasets sourced from different geographical locations, each of which contains varying percentages of missing values. The results of our investigation revealed that, in many cases, the integration of the proposed Pseudo GPHS Labeling on top of eight diverse supervised learning algorithms can significantly enhance the quality of results, highlighting its effectiveness in improving the overall performance of the machine learning models. Besides, the results emphasize the potential of Random Forest and eXtreme Gradient Boosting as the best model to integrate with the Pseudo GPHS Labeling approach. Finally, the investigation using interpretability tools highlights the importance of the growing degree days (GDD) as an input feature, while also allowing for a visualization of the improvement obtained by using the proposed approach.

In summary, this study has three primary contributions. First, it develops a cost-effective method for predicting GPHS using machine learning with environmental and climatic data obtained in a non-intrusive way, making a significant contribution to this underexplored area. Second, it illustrates the efficacy of pseudo-labeling as a semi-supervised technique in enhancing predictions by mitigating missing target values, thereby advancing the precision and comprehensiveness of GPHS prediction. Through an extensive evaluation involving eight machine learning algorithms and seven datasets, the proposed approach consistently achieves significant improvements in GPHS estimation. Finally, to the best of our knowledge, our study is one of the first in viticulture to utilize explainable artificial intelligence techniques for interpreting and explaining the obtained results.

2. Methodology

2.1. Data

The phenological stage dataset was collected from six vineyards of six different farm companies in the Friuli Venezia Giulia region, Italy (Table 1). Data was collected during the vegetative season (April–September) from 2009 to 2022. All vineyards employed the Guyot system, with a plant spacing of 0.9–1.0 m within rows and rows spaced 2.5–2.6 m apart. At pruning time, 8–9 buds per shoot were retained annually, and plant shoots were managed following the vertical shoot positioning (VSP) criteria. Phenology observations were conducted annually in four representative plots within each vineyard. For every plot, three plants were chosen, and their phenological stages were evaluated using the BBCH scale, as described by (Meier, 2018) and detailed in Table 2. The mean BBCH value was calculated for the three plants within each plot, and an overall average was determined across the four plots in each vineyard. Phenology monitoring occurred weekly, with increased frequency (twice weekly) during the grapevine moth development phase for pest monitoring purposes. The study focused on five stages: stage 5 (inflorescence emergence, covering BBCH codes 53, 55, and 57), stage 6 (flowering, covering BBCH codes from 60 to 69), stage 7 (development of fruits, BBCH codes 71, 73, 75, 77, and 79), stage 8 (ripening of berries, covering codes 81, 83, 85, and 89), and stage 9 (senescence), only code 91 (post-harvest; end of wood maturation). Table 3 provides a statistical summary, displaying descriptive statistics for the datasets.

The phenological data was linked to meteorological measurements collected from weather stations located within the vineyard areas. Weather station data included seven attributes: average daily temperature (°C) (Tmed), cumulative daily rainfall (mm) (CDR), maximum daily temperature (°C) (Tmax), minimum daily temperature (°C) (Tmin), and the presence of days with maximum temperatures above 30 (D30) and 35 (D35) degrees Celsius. The average temperature was used to calculate growing degree days (GDD), which were determined as the cumulative temperature exceeding a base threshold of 10 °C, established as the growth baseline for grapevines in the measured area

Table 1
Details of all the vineyard positioning, and cultivars. All the farms were in Friuli Venezia Giulia region, Italy.

Farm	Location	Cultivars	Degree North	Degree East-West	Abbreviation for vineyard
Monviert	Cividale del Friuli	Ribolla gialla	46,05626	13,45,445	SM
Vigna Petrussa	Prepotto	Schioppettino	46,05012	13,49,074	PV
Ermacora	Premariacco	Sauvignon blanc	46,03941	13,41,498	IES
Livio Felluga	Manzano	Tocai friulano	46,00609	13,42,275	MR
Ermacora	Premariacco	Pinot Grigio	46,03965	13,41,530	IEP
Jacuss	Torreano	Pinot Bianco	46,11,438	13,40,398	CJ

(Caffarra and Eccel, 2010). Table 4 shows each dataset's total number of records with and without labels. After merging all datasets, we created a comprehensive dataset named 'Merge Dataset,' which facilitates thorough analysis despite missing values.

2.2. The proposed approach

We propose a semi-supervised approach that uses pseudo-labeling to boost GPHS prediction based on climatic data. This approach allows us to use datasets that present many unlabeled samples and only a few GPHS-labeled ones, as shown in Fig. 1 (left). Fig. 1 presents an overview of the proposed approach, which consists of three main phases. In the first phase, an initial model is trained using supervised learning on the labeled subset of the original dataset, which contains GPHS labels provided by an expert (colored in orange in the Figure). In the second phase, referred to as Pseudo GPHS Labeling, the previously trained supervised model is used to generate pseudo-labels (colored in blue in the Figure) for the unlabeled subset of the data. Integrating the original labeled data with newly pseudo-labeled data can greatly enlarge the dataset available for supervised learning, providing a valuable resource for enhancing predictive GPHS models. Finally, in the third phase, a final supervised model is trained on the merged dataset, leveraging the newly obtained augmented dataset to further improve performance. Explainable artificial intelligence techniques are also integrated into the methodology to provide explanations on the predictions made by the model, which is useful to gain further insights from domain experts. The following section describes in detail the proposed approach and highlights its inherent generality, yielding a methodology applicable to various machine learning models, ultimately resulting in enhanced GPHS prediction.

Table 2
Our study considers the phenological growth stages and BBCH-identification keys of grapes.

Growth stage	Code	Description
5: Inflorescence emerge	53	Inflorescences visible
	55	Inflorescences swelling, flowers closely pressed together
	57	Inflorescences fully developed; flowers separating
6: Flowering	60	First flower hoods detached from the receptacle
	61	Beginning of flowering: 10 % of flower hoods fallen
	62	20 % of flower hoods fallen
	63	Early flowering: 30 % of flower hoods fallen
	64	40 % of flower hoods fallen
	65	Full flowering: 50 % of flower hoods fallen
	66	60 % of flower hoods fallen
	67	70 % of flower hoods fallen
	68	80 % of flower hoods fallen
	69	End of flowering
7: Development of fruits	71	Fruit set: Young fruits begin to swell and remains of flowers are lost
	73	Berries groat-sized, bunches begin to hang
	75	Berries pea-sized, bunches hang
	77	Berries beginning to touch
	79	The majority of berries touching
8: Ripening of berries	81	Beginning of ripening: berries begin to develop variety-specific color
	83	Berries developing color
	85	Softening of berries
	89	Berries ripe for harvest
9: Senescence	91	After harvest; the end of wood maturation

2.2.1. Pseudo GPHS labeling

In the proposed Pseudo GPHS Labeling strategy, in the first step, a model is initially trained using supervised learning with a machine learning algorithm and a limited subset of labeled examples from the original dataset. Each labeled example corresponds to one of the GPHS categories based on our datasets and the phenological stage evaluated by an expert in the field. Notably, any learning algorithm is suitable for the proposed approach, emphasizing its general nature (see Section 2.2.2 for more details). In our case study, we relied on regression models to capture the continuous nature of plant growth and adapt to climate-driven variability, as supported by studies such as (Chuine and Régnière, 2017) and (Parmesan and Yohe, 2003). These models are particularly effective for predicting the progression of plant development in response to environmental changes, such as temperature, sunlight, and humidity. Regression models ensure accurate predictions across diverse climatic conditions by accommodating complex, nonlinear relationships between ecological variables and plant phenology (Fu et al., 2020). Additionally, by mapping continuous outputs to the nearest discrete BBCH stages, the model translates detailed growth dynamics into actionable insights, seamlessly bridging scientific precision with practical application (Schieler et al., 2024).

In the second step of the proposed strategy, the initial supervised model is subsequently deployed to make predictions on a more extensive pool of unlabeled data. The predictions generated by the model are treated as Pseudo-labels, effectively assigning class labels to previously unlabeled data points based on the model's output (Grandvalet and Bengio, 2004; Lee, 2013). Finally, these pseudo-labeled examples are then used alongside the original labeled data to train a new model for final prediction, as shown in step 3 of Fig. 1. By following this procedure, it becomes possible to effectively use a larger amount of training data without requiring expert annotations on all the data points, resulting in increased annotation efficiency and convenience. Moreover, the additional information provided by the pseudo-labeled data leads to improved accuracy and better generalization capabilities of the model.

The Pseudo GPHS Labeling strategy can be described as follows:

Input: the supervised learning algorithm A, the set of labeled examples X_l, the set of unlabeled examples X_u, the total number of iterations I.

Output: the model f_i trained using A on both X_l and X_u after I iterations.

Table 3
Statistics related to the distribution of target variables in the data. The table reports the average value (avg), standard deviation (std), minimum (min), maximum (max), 25th, 50th, and 75th percentiles (Q1, Q2, Q3).

Dataset	AVG	Std	Min	Max	Q1	Q2	Q3
SM	75.70	9.83	53	91	73	79	83
PV	76.01	10.56	53	91	74	77	84
IES	74.06	9.35	55	91	75	75	82
MR	77.73	11.00	53	91	75	79	85
IEP	75.85	9.95	53	89	74	79	84
CJ	75.70	9.83	53	91	73	79	83

Table 4
Total number of records, number of records with and without phenological stage labels.

Dataset	Tot Num of record	Num of record with label	Num of record without label
SM	5114	539	4575
PV	5113	606	4507
IES	5108	552	4556
MR	5130	587	4543
IEP	5112	537	4575
CJ	5115	559	4556
Merge	30,692	3380	27,312

1. Model training using the supervised algorithm A on the labeled data X_l for GPHS prediction, obtaining the initial model $f_{i=0}$.
2. Pseudo-label generation according to Eq. 1, where $\tilde{x}_u^{(i)}$ represents the Pseudo-labels generated by the model f_i for the unlabeled dataset X_u at iteration i.

$$\tilde{x}_u^{(i)} = f_i(X_u) \tag{1}$$

3. Combination of the labeled and Pseudo-label data according to Eq. 2, where $X_{combined}^{(i)}$ is the combined dataset.

$$X_{combined}^{(i)} = X_l \cup \tilde{x}_u^{(i)} \tag{2}$$

4. Model training using algorithm A on the combined dataset $X_{combined}^{(i)}$ for GPHS prediction, obtaining the model f_{i+1}.
5. Moving to next iteration (i + 1) from step 2, until i = I.
6. Finally, f_I represents the final model obtained with the Pseudo GPHS labeling strategy.

In our scenario, we consider $I = 1$. For algorithm A, any supervised machine learning technique is viable. Hence, this study integrates the proposed Pseudo GPHS Labeling strategy with eight diverse supervised learning models (see Appendix A), further highlighting its general nature and providing evidence of the effectiveness in advancing the accuracy of GPHS prediction based on climatic data (Section 3). Similar approaches based on pseudo-labels also found success in various applications, from image classification to natural language processing, where labeled data is scarce or expensive to obtain, offering an efficient means to harness the potential of unlabeled datasets (Berrebbi et al., 2022). By seamlessly combining Pseudo GPHS labeling with the range of models detailed below, the research explores the synergies and unique strengths of each algorithm within this general framework. By using the proposed Pseudo GPHS Labeling strategy, it is possible to mitigate overfitting, enhance generalization, and improve overall predictive performance. Through a comparative analysis, this study aims to uncover the most effective combination, offering valuable insights into the optimal utilization of Pseudo GPHS Labeling for precise and reliable grapevine phenology prediction.

The proposed Pseudo GPHS Labeling strategy is versatile and can be applied to any machine learning algorithm. In this research, we evaluated eight models: Multiple Linear Regression (MLR), Random Forest (RF), Gradient Boosting Decision Tree (GBDT), eXtreme Gradient Boosting (XGBoost), Categorical Boosting (CatBoost), Multilayer Perceptron (MLP), Support Vector Regression (SVR), and K-Nearest Neighbors (KNN). Results were generated based on these models, with definitions for each provided in Appendix A.

3. Experimental setup

To enhance both model learning and robust performance evaluation, a 5-fold cross-validation strategy was implemented. This approach effectively reduces the impact of data randomness and promotes increased diversity in the training sets. The dataset was partitioned into five equal subsets or "folds," with each iteration involving training on four folds (80 % of the data) and testing on the remaining one-fold (20 % of the data). This process was iterated five times, with each fold serving as the test set once. The computed average performance metrics across all iterations and runs provided a more reliable evaluation of the model's effectiveness. In the context of mitigating overfitting, 5-fold cross-validation indirectly addresses this concern by assessing the model's generalization across different subsets of the data. This ensures that the model's performance is not excessively tailored to specific training data, helping to identify and prevent overfitting issues.

All experiments were executed on a dedicated high-performance computing cluster featuring dual Intel Xeon processors, each with 48 CPU cores and 256 GB of RAM. We harnessed NVIDIA Tesla V100 GPUs with 32 GB of VRAM for GPU-accelerated tasks, powered by the CUDA framework. Operating under Ubuntu Linux (Python version 3.8), our research leveraged the TensorFlow framework (v2.3.0) and Keras for model development and training, while custom Python scripts, along with libraries like NumPy and OpenCV, handled data preprocessing and augmentation.

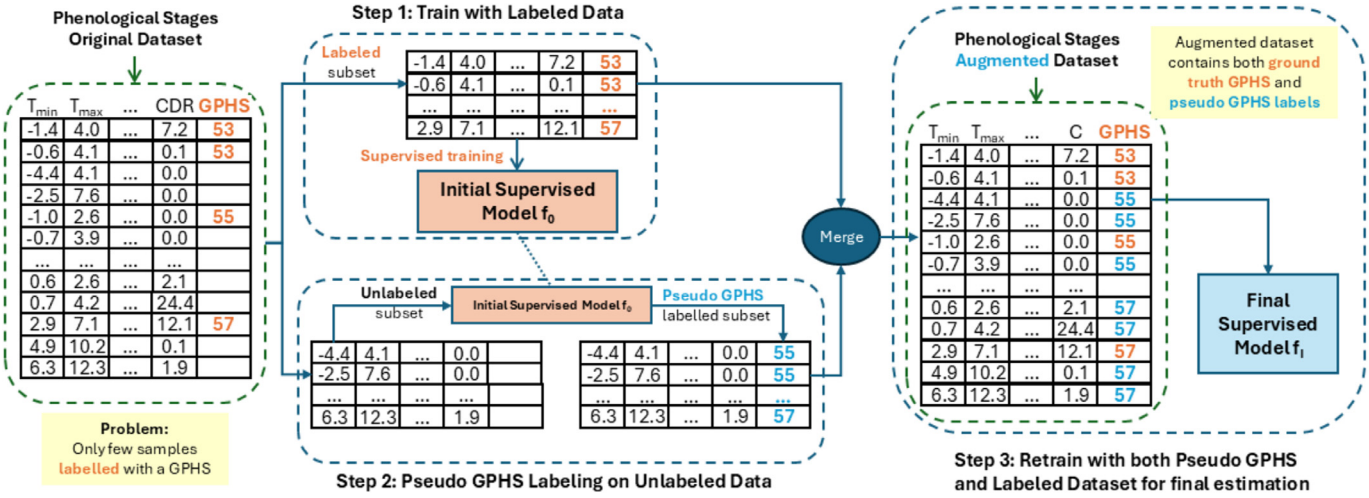


Fig. 1. An overview of the proposed general framework, called the Pseudo GPHS Labeling strategy.

Table 5
Hyperparameter overview: definitions and grid search ranges for various algorithms.

Models	Parameter name	Meaning	Range
CatBoost	iterations	Max count of trees.	[100, 150, 200, 250]
	learning_rate	Boosting learning rate	[0.03, 0.1]
	depth	Depth of a tree.	[2, 4, 6, 8, 10]
	l2_leaf_reg	Coefficient at the L2 regularization term of the cost function.	[0.2, 0.5, 1, 3, 4]
GBDT	n_estimators	The number of boosting stages to perform.	[1, 2, 5, 10, 20, 50, 100, 200, 500, 1000, 2000]
	max_leaf_nodes	Grow trees with max_leaf_nodes in the best-first fashion.	[2, 5, 10, 20, 50, 100]
	learning_rate	Boosting learning rate	[1, 0.1, 0.01,0.001]
	max_depth	Maximum depth of the individual regression estimators	[1,2,4]
	subsample	The fraction of samples to be used for fitting the individual base learners.	[0.5, 0.75, 1]
XGBoost	max_depth	Maximum tree depth for base learners	[4, 5, 10]
	n_estimators	The number of boosting stages to perform.	[500, 600, 700, 800]
	learning_rate	Boosting learning rate	[0.01, 0.015, 0.02]
RF	n_estimators	The number of trees in the forest.	[200, 400, 600, 1000, 1200]
	min_samples_split	The minimum number of samples required to split an internal node	[10, 12, 14, 16]
	min_samples_leaf	The minimum number of samples required to be at a leaf node.	[4, 5, 6, 8]
	max_features	The number of features to consider when looking for the best split.	[4, 5, 6]
	max_depth	The maximum depth of the tree.	[100, 110, 120, 130, 140, 150]
MLP	max_iter	Number of training epochs	[10, 20, 50]
	learning_rate_init	Initial learning rate	[0.01, 0.10]
	hidden_layer_sizes	Number of neurons in the hidden layer	[2, 4, 8, 64, 128]
KNN	n_neighbors	The number of neighbors.	[1, 2, …, 30]
SVR	C	Regularization parameter.	[1000, 100, 10, 1, 0.01]
	gamma	Kernel coefficient.	[1e-5, 1e-4, 1e-3, 1e-2]
MLR	–		

3.1. Hyperparameters tuning

In our study, we aimed to optimize the performance of our models and refine our estimates for GPHS. To carry out the hyperparameters tuning, we adopted the grid search technique. This method involves constructing a grid of hyperparameter values according to literature and systematically training the models using every conceivable combination within that grid. We considered k-fold cross-validation within our grid search method, allowing us to specify the number of folds in cross-validation. Specifically, we utilized all available data and implemented 5-fold cross-validation to assess each combination and determine the best set of parameters. Table 5 provides a comprehensive overview of the hyperparameters we considered for each model, their significance, and the range of values explored during the search process.

3.2. Evaluation metrics

For the evaluation of our models, we employ the two most common error metrics: the coefficient of determination (R^2), and root-mean-square error (RMSE). Typically, lower values of RMSE indicate superior performance, while a higher R^2 signifies better estimation accuracy. Eqs. 1 and 2 define R^2 and RMSE, respectively. In these equations, $\hat{y}$ represents the predicted value, y_i stands for the observed measured value, $\bar{y}$ denotes the meaning of the observed values, and n represents the size of the test sample.

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y})^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \quad (1)$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y})^2} \quad (2)$$

4. Results

4.1. Initial results obtained with standard supervised learning approaches

In our initial exploration using standard supervised learning approaches, we encountered a diverse landscape of predictive performance across various models and datasets (Table 6). These approaches, while well-established, demonstrated varying degrees of effectiveness in capturing the intricate patterns of GPHS.

For instance, CatBoost exhibited promising results in certain datasets such as 'IEP', where it achieved a commendable R^2 of 0.96 ± 0.00. However, despite this relatively high correlation coefficient, the RMSE of 1.85 ± 0.10 suggested a notable level of residual variability in its predictions, indicating potential for enhanced predictive precision. Similarly, GBDT showcased encouraging performance, particularly in the 'MR' dataset, where it achieved an R^2 of 0.96 ± 0.01. This signifies a strong ability to explain the variance in GPHS predictions, yet its RMSE of 1.59 ± 0.19 suggests room for refinement in minimizing prediction errors.

On the other hand, different machine learning models, such as KNN and RF, showed moderate performance across various datasets. While KNN exhibited relatively high R^2 values, indicating a good correlation with actual data, its comparatively higher RMSE values suggest room for improvement. MLR and SVR models had mixed results, with MLR showing lower R^2 values and higher RMSE in some datasets, indicating difficulty in capturing nonlinear relationships. SVR had higher R^2 values in certain datasets but struggled with elevated RMSE, suggesting discrepancies between predicted and actual outcomes. In summary, while standard supervised learning approaches provided valuable insights into GPHS prediction, their performance was accompanied by certain limitations, including residual errors and challenges in capturing complex nonlinear relationships within the data. These findings underscore the need for innovative methodologies to enhance predictive accuracy and address the intricacies of viticultural datasets.

4.2. Results obtained with the proposed pseudo GPHS labeling approach

Implementing our novel Pseudo GPHS Labeling approach yielded substantial advancements in predictive accuracy across various datasets and models, underscoring its robustness and versatility within viticulture (See Table 7). In the CJ dataset, for instance, the XGB model emerged as a standout performer, showcasing the most significant improvement. With the R^2 score soaring from 0.09 to 0.80 and the RMSE plummeting from 9.15 to 2.55. Similarly, in the "IEP dataset," the RF model demonstrated remarkable improvements, with the R^2 score surging from 0.95 to 0.98 and the RMSE declining from 1.65 to 0.32.

Table 6
Performance of the models considered in the supervised learning approach for the target variable. The reported metrics are R^2 (higher is better) and RMSE (lower is better). The best result for each Dataset is bolded.

Model	Metric	CJ	IEP	IES	MR	PV	SM	Merge
Cat	R^2	**0.15 ± 0.10**	**0.96 ± 0.00**	0.95 ± 0.01	0.95 ± 0.00	0.28 ± 0.06	0.95 ± 0.02	**0.10 ± 0.01**
Boost	RMSE	**8.86 ± 0.47**	**1.85 ± 0.10**	1.61 ± 0.38	1.89 ± 0.14	7.96 ± 0.45	1.56 ± 0.47	**9.73 ± 0.22**
GBDT	R^2	0.15 ± 0.07	0.96 ± 0.00	0.94 ± 0.01	**0.96 ± 0.01**	0.23 ± 0.04	0.96 ± 0.02	0.09 ± 0.01
	RMSE	8.87 ± 0.34	1.64 ± 0.11	1.54 ± 0.47	**1.59 ± 0.19**	8.24 ± 0.40	1.47 ± 0.49	9.74 ± 0.24
KNN	R^2	0.13 ± 0.11	0.95 ± 0.00	0.94 ± 0.01	**0.96 ± 0.01**	0.27 ± 0.06	0.96 ± 0.02	0.02 ± 0.01
	RMSE	8.98 ± 0.47	1.90 ± 0.11	1.50 ± 0.46	**1.58 ± 0.37**	8.01 ± 0.46	1.40 ± 0.48	10.14 ± 0.22
MLR	R^2	0.08 ± 0.06	0.83 ± 0.04	0.87 ± 0.03	0.89 ± 0.02	0.14 ± 0.04	0.87 ± 0.03	0.05 ± 0.01
	RMSE	9.23 ± 0.40	3.86 ± 0.20	3.87 ± 0.40	3.47 ± 0.32	8.69 ± 0.33	3.46 ± 0.39	9.96 ± 0.21
MLP	R^2	0.66 ± 0.32	0.33 ± 0.14	0.57 ± 0.10	0.60 ± 0.17	0.22 ± 0.14	0.61 ± 0.26	0.00 ± 0.01
	RMSE	12.34 ± 0.90	7.58 ± 0.66	7.05 ± 0.80	6.54 ± 1.42	10.36 ± 0.60	5.97 ± 1.87	10.23 ± 0.24
RF	R^2	0.14 ± 0.13	0.95 ± 0.00	0.95 ± 0.01	0.96 ± 0.01	**0.29 ± 0.06**	0.96 ± 0.02	0.04 ± 0.01
	RMSE	8.92 ± 0.55	1.65 ± 0.08	1.55 ± 0.45	1.72 ± 0.31	**7.93 ± 0.50**	1.51 ± 0.47	10.00 ± 0.23
SVR	R^2	0.04 ± 0.16	0.94 ± 0.01	**0.95 ± 0.00**	0.96 ± 0.01	0.16 ± 0.10	**0.95 ± 0.01**	0.06 ± 0.01
	RMSE	9.38 ± 0.66	1.10 ± 0.15	**1.40 ± 0.16**	1.67 ± 0.21	8.61 ± 0.69	**1.34 ± 0.21**	10.51 ± 0.24
XGB	R^2	0.09 ± 0.14	0.95 ± 0.01	0.98 ± 0.01	0.96 ± 0.01	0.22 ± 0.07	0.94 ± 0.03	0.08 ± 0.01
	RMSE	9.15 ± 0.47	1.18 ± 0.16	1.55 ± 0.47	1.67 ± 0.39	8.29 ± 0.62	1.83 ± 0.74	9.83 ± 0.23

Moreover, in the "IES dataset," our approach led to significant improvements across various models, with the RF model exhibiting notable enhancements. The R^2 score ranged from 0.95 to 0.98, while the RMSE showed a decrease from 1.55 to 0.46, indicating the robustness of our methodology in addressing data sparsity challenges and enhancing predictive accuracy. Across all datasets, including "MR," "PV," "SM," and "Merge," consistent patterns of improvement were observed, with the RF model consistently outperforming others. While the performance on the Merge dataset is lower than on the individual datasets, the decline in predictive performance can be attributed to several factors. Data heterogeneity across locations, driven by variations in climate, microclimates, and management practices, makes it challenging for the model to generalize effectively. Differences in rainfall, temperature fluctuations, and cultivar-specific responses—such as those of Ribolla Gialla, Schioppettino, and Pinot Grigio—further complicate predictions, as each grape variety exhibits unique phenological behavior and reacts differently to climatic variables. Additionally, the geographical and climatic variability within the Friuli Venezia Giulia region, including differences in latitude, altitude, and vineyard orientation, creates distinct growing conditions that a unified model struggles to capture accurately. Nonetheless, the addition of the proposed Pseudo GPHS Labeling strategy is helpful in all the considered models and datasets, with substantial improvements even on the Merge dataset, improving the average RMSE from 10.01 to about 3.28.

Generally, our comprehensive analysis underscores the effectiveness and reliability of the pseudo-GPHS labeling approach in enhancing predictive accuracy in viticulture. By addressing data sparsity challenges and effectively leveraging unlabeled data, our methodology offers a promising solution for improving predictive modeling and informing sustainable crop management practices in viticulture. Besides, these updated results emphasize the potential of XGB and RF as robust tools to integrate with the Pseudo-labeling approach for handling missing value and GPHS prediction, contributing to the advancement of viticulture research and practices.

In addition, we applied our approach and evaluated the performance of different machine learning models in predicting GPHS across different phenological phases (Table 8). Among these models, CatBoost demonstrated the highest overall performance, consistently achieving the best balance between high R^2 values and low RMSE across various datasets and phenological stages. For example, during the Inflorescence Emergence phase, CatBoost attained $R^2 = 0.97$ with an RMSE of 0.10 in the IEP dataset, outperforming other models in this phase. Similarly, in the Development of Fruits phase, CatBoost achieved $R^2 = 0.96$ with an RMSE of 0.46 in the MR dataset, indicating strong predictive capability with minimal error. These results highlight CatBoost's robustness in handling complex relationships within the data, making it particularly effective for GPHS prediction.

Following CatBoost, XGBoost emerged as the second-best performing model, often achieving comparable results and, in some cases, outperforming CatBoost in certain datasets. For instance, in the Development of Fruits phase, XGBoost achieved the highest R^2 (0.97) and the lowest RMSE (0.33) in the IES dataset, demonstrating its ability to capture patterns efficiently. Moreover, in the Veraison phase, XGBoost maintained a competitive R^2 (~0.94–0.97) across multiple datasets, with a particularly strong performance in the MR dataset ($R^2 = 0.97$, RMSE = 0.33). Despite these strengths, XGBoost exhibited slightly

Table 7
Performance of the considered models in semi-supervised learning approach for the target variable. The reported metrics are R^2 (higher is better) and RMSE (lower is better). The best result for each Dataset is bolded.

Model	Metric	CJ	IEP	IES	MR	PV	SM	Merge
Cat	R^2	0.63 ± 0.05	0.98 ± 0.03	0.98 ± 0.00	0.98 ± 0.00	0.67 ± 0.03	0.98 ± 0.00	0.27 ± 0.01
Boost	RMSE	2.85 ± 0.19	0.48 ± 0.04	0.57 ± 0.02	0.65 ± 0.02	2.73 ± 0.15	0.54 ± 0.05	3.22 ± 0.10
GBDT	R^2	0.62 ± 0.04	0.97 ± 0.00	0.97 ± 0.00	0.97 ± 0.00	0.70 ± 0.03	0.66 ± 0.02	0.34 ± 0.02
	RMSE	2.86 ± 0.17	1.31 ± 0.04	0.84 ± 0.04	1.13 ± 0.14	2.73 ± 0.18	0.55 ± 0.00	3.23 ± 0.10
KNN	R^2	0.69 ± 0.04	0.98 ± 0.01	0.98 ± 0.00	0.98 ± 0.06	0.76 ± 0.02	0.98 ± 0.00	0.38 ± 0.02
	RMSE	2.89 ± 0.19	0.35 ± 0.04	0.51 ± 0.13	0.59 ± 0.18	2.71 ± 0.14	0.53 ± 0.18	3.36 ± 0.10
MLR	R^2	0.61 ± 0.02	0.97 ± 0.00	0.97 ± 0.00	0.61 ± 0.02	0.61 ± 0.02	0.97 ± 0.00	0.25 ± 0.02
	RMSE	3.01 ± 0.10	1.26 ± 0.08	1.28 ± 0.08	2.93 ± 0.10	2.93 ± 0.10	1.22 ± 0.12	3.29 ± 0.11
MLP	R^2	0.73 ± 0.07	0.96 ± 0.00	0.97 ± 0.00	0.97 ± 0.00	0.56 ± 0.10	0.96 ± 0.02	0.61 ± 0.2
	RMSE	4.03 ± 0.51	2.82 ± 0.14	2.83 ± 0.22	2.34 ± 0.26	3.71 ± 0.41	2.61 ± 1.15	3.37 ± 0.1
RF	R^2	0.77 ± 0.05	**0.98 ± 0.00**	**0.98 ± 0.00**	**0.98 ± 0.00**	0.80 ± 0.03	**0.98 ± 0.00**	**0.53 ± 0.02**
	RMSE	2.65 ± 0.25	**0.32 ± 0.04**	**0.46 ± 0.11**	**0.55 ± 0.10**	2.53 ± 0.16	**0.53 ± 0.15**	**3.14 ± 0.11**
SVR	R^2	0.01 ± 0.01	0.98 ± 0.00	0.98 ± 0.00	0.98 ± 0.00	0.04 ± 0.00	0.98 ± 0.00	0.61 ± 0.15
	RMSE	3.35 ± 0.23	0.43 ± 0.05	0.50 ± 0.05	0.61 ± 0.10	3.17 ± 0.17	0.55 ± 0.12	3.50 ± 0.11
XGB	R^2	**0.80 ± 0.07**	**0.98 ± 0.00**	**0.98 ± 0.00**	**0.98 ± 0.00**	**0.82 ± 0.02**	**0.98 ± 0.00**	0.47 ± 0.02
	RMSE	**2.55 ± 0.24**	**0.33 ± 0.04**	**0.39 ± 0.02**	**0.47 ± 0.01**	**2.48 ± 0.13**	**0.41 ± 0.03**	3.16 ± 0.11

Table 8
Performance of the considered models in a semi-supervised learning approach for predicting the GPHS variable, analyzed separately by phase. The reported metrics are R^2 (higher is better) and RMSE (lower is better). The best result for each dataset is highlighted in bold.

Stage	Datasets	Cat Boost		GBDT		RF		XGB	
		R^2	RMSE	R^2	RMSE	R^2	RMSE	R^2	RMSE
Development of fruits	CJ	0.89 ± 0.03	0.58 ± 0.05	0.88 ± 0.02	0.61 ± 0.05	0.88 ± 0.02	0.61 ± 0.05	**0.90 ± 0.03**	**0.57 ± 0.07**
	IEP	0.96 ± 0.12	0.15 ± 0.01	0.97 ± 0.10	0.13 ± 0.01	**0.98 ± 0.06**	**0.11 ± 0.05**	0.96 ± 0.20	0.13 ± 0.07
	PV	**0.75 ± 0.02**	**1.02 ± 0.04**	0.74 ± 0.02	1.04 ± 0.03	**0.75 ± 0.02**	**1.01 ± 0.03**	**0.75 ± 0.32**	**1.03 ± 0.04**
	MR	0.96 ± 0.01	0.46 ± 0.08	0.96 ± 0.01	0.47 ± 0.08	**0.97 ± 0.01**	**0.42 ± 0.09**	0.96 ± 0.01	0.50 ± 0.09
	IES	0.90 ± 0.05	0.62 ± 0.12	0.90 ± 0.05	0.62 ± 0.11	0.91 ± 0.05	0.59 ± 0.13	**0.97 ± 0.01**	**0.33 ± 0.04**
	SM	0.82 ± 0.05	0.91 ± 0.12	0.82 ± 0.06	0.93 ± 0.13	**0.83 ± 0.05**	**0.90 ± 0.12**	0.82 ± 0.06	0.93 ± 0.13
	Merge	0.62 ± 0.01	1.17 ± 0.01	0.56 ± 0.01	1.26 ± 0.02	**0.68 ± 0.01**	**1.07 ± 0.02**	0.66 ± 0.01	1.10 ± 0.02
Veraison	CJ	**0.75 ± 0.20**	**0.64 ± 0.32**	0.74 ± 0.10	0.50 ± 0.18	0.73 ± 0.09	1.14 ± 0.02	0.74 ± 0.05	0.51 ± 0.04
	IEP	0.97 ± 0.01	0.31 ± 0.04	0.95 ± 0.01	0.39 ± 0.04	0.96 ± 0.01	0.35 ± 0.04	**0.97 ± 0.01**	**0.33 ± 0.04**
	IES	**0.94 ± 0.02**	**0.62 ± 0.11**	0.92 ± 0.02	0.70 ± 0.10	0.94 ± 0.03	0.61 ± 0.08	0.94 ± 0.02	0.58 ± 0.12
	MR	**0.97 ± 0.01**	**0.31 ± 0.04**	0.95 ± 0.01	0.39 ± 0.04	0.96 ± 0.11	0.34 ± 0.05	**0.97 ± 0.01**	**0.33 ± 0.04**
	SM	**0.91 ± 0.05**	**0.20 ± 0.06**	0.89 ± 0.05	0.22 ± 0.05	0.90 ± 0.05	0.29 ± 0.01	0.90 ± 0.05	0.21 ± 0.06
	Merge	0.78 ± 0.03	0.75 ± 0.06	0.75 ± 0.03	0.79 ± 0.06	**0.78 ± 0.05**	**0.71 ± 0.05**	0.77 ± 0.04	0.76 ± 0.07
Inflorescence emerge	CJ	0.49 ± 0.40	0.29 ± 0.25	0.28 ± 0.24	0.42 ± 0.28	0.42 ± 0.36	0.37 ± 0.25	**0.51 ± 0.39**	**0.31 ± 0.16**
	IEP	**0.97 ± 0.02**	**0.10 ± 0.04**	0.95 ± 0.02	0.15 ± 0.04	0.96 ± 0.03	0.11 ± 0.05	0.96 ± 0.02	0.12 ± 0.03
	IES	0.88 ± 0.03	0.30 ± 0.03	0.86 ± 0.03	0.32 ± 0.04	**0.88 ± 0.04**	**0.29 ± 0.05**	0.86 ± 0.03	0.31 ± 0.04
	MR	**0.97 ± 0.02**	**0.10 ± 0.04**	0.97 ± 0.02	0.11 ± 0.04	0.96 ± 0.01	0.13 ± 0.03	0.96 ± 0.02	0.12 ± 0.03
	SM	**0.89 ± 0.03**	**0.29 ± 0.04**	0.89 ± 0.03	0.30 ± 0.05	0.89 ± 0.03	0.30 ± 0.05	0.88 ± 0.04	0.31 ± 0.05
	Merge	**0.85 ± 0.04**	**0.43 ± 0.05**	0.82 ± 0.05	0.48 ± 0.05	0.84 ± 0.06	0.47 ± 0.01	0.83 ± 0.03	0.46 ± 0.06
Flowering	CJ	0.49 ± 0.06	0.81 ± 0.02	0.47 ± 0.09	0.81 ± 0.02	**0.51 ± 0.74**	**0.61 ± 0.30**	0.41 ± 0.42	0.73 ± 0.11
	IEP	0.96 ± 0.01	0.46 ± 0.08	0.96 ± 0.01	0.47 ± 0.08	**0.97 ± 0.01**	**0.42 ± 0.09**	0.96 ± 0.01	0.50 ± 0.09
	IES	**0.84 ± 0.05**	**0.46 ± 0.10**	0.84 ± 0.03	0.47 ± 0.03	0.82 ± 0.01	0.48 ± 0.04	**0.84 ± 0.08**	**0.46 ± 0.07**
	MR	0.80 ± 0.22	0.50 ± 0.32	0.77 ± 0.22	0.54 ± 0.32	0.76 ± 0.15	0.60 ± 0.18	**0.96 ± 0.01**	**0.50 ± 0.09**
	SM	**0.75 ± 0.02**	**1.02 ± 0.04**	0.74 ± 0.02	1.04 ± 0.03	0.74 ± 0.05	1.51 ± 0.04	0.73 ± 0.01	1.95 ± 0.07
	Merge	0.50 ± 0.54	0.68 ± 0.20	0.43 ± 0.46	0.75 ± 0.17	**0.52 ± 0.57**	**0.65 ± 0.21**	0.48 ± 0.56	0.68 ± 0.20

higher variability in some datasets, particularly in the Merge dataset, where RMSE values fluctuated more than those observed for CatBoost.

Overall, the results indicate that CatBoost is the most reliable model for predicting GPHS, consistently delivering the most stable and accurate predictions across different datasets and phenological phases. XGBoost, while also a strong contender, exhibited slightly more variability, making it more sensitive to dataset-specific conditions. The best results across the models were consistently produced for the IEP and MR datasets, with R^2 values exceeding 0.95, while the lowest predictive performance was obtained in the CJ and Merge datasets, with R^2 values as low as 0.41 and higher RMSE, particularly in the Flowering phase. Across all phenological stages, the Development of Fruits and Veraison phases showed the most stable performance, while the Flowering phase exhibited the highest variability, with substantial performance fluctuations across models. These findings emphasize the impact of dataset characteristics and phenological phases on model performance and suggest that further refinements, such as additional feature engineering or data augmentation, may be required for the lower-performing datasets.

4.3. Results obtained with the proposed approach for day of year (DOY)

The Pseudo GPHS Labeling method was employed to predict the Day of Year (DOY) for grapevine growth stages, focusing on four key stages: stage 5 (inflorescence emergence, BBCH codes 53, 55, and 57), stage 6 (flowering, BBCH codes 60–69), stage 7 (development of fruits, BBCH codes 71, 73, 75, 77, and 79), and stage 8 (ripening of berries, BBCH codes 81, 83, 85, and 89). The predictive outcomes are summarized in Table 9, highlighting the performance of multiple machine learning models under various datasets and conditions.

The combination of the Pseudo Labeling method and CatBoost emerged as the top-performing model, achieving the highest predictive accuracy across multiple datasets and growth stages. For flowering, CatBoost achieved the highest R^2 (0.94 ± 0.01) and lowest RMSE (6.41 ± 0.30) on the IEP dataset. Similarly, for ripening, CatBoost delivered competitive performance with R^2 values of 0.94 ± 0.01 for IEP and 0.73 ± 0.07 for IES, alongside RMSE values of 6.41 ± 0.30 for IEP and 6.21 ± 0.86 for IES. For the development of fruits (stage 7), CatBoost consistently demonstrated high accuracy on the "Merge" dataset, with an R^2 of 0.99 ± 0.00 and RMSE of 7.17 ± 0.79, outperforming other models. For inflorescence emergence (stage 5), CatBoost maintained strong performance, achieving $R^2 = 0.73 \pm 0.10$ and RMSE = 16.04 ± 2.01 for the CJ dataset. Furthermore, GBDT also showcased strong predictive ability, particularly for specific datasets and stages. For instance, it matched CatBoost in flowering predictions on the IEP dataset with $R^2 = 0.94 \pm 0.01$ and RMSE of 6.43 ± 0.47. It also performed well for ripening on IES, achieving $R^2 = 0.75 \pm 0.07$ and RMSE of 6.51 ± 0.97. However, GBDT occasionally exhibited higher variability in RMSE compared to CatBoost. In contrast, RF was less consistent. RF demonstrated competitive R^2 values in certain scenarios but often suffered from higher RMSE, such as in flowering predictions on the IEP dataset, where RMSE was 6.54 ± 0.42. KNN and MLR, meanwhile, showed considerably lower performance, particularly for ripening predictions, where their R^2 values were significantly lower, and RMSE values were substantially higher, suggesting limited applicability in complex datasets. In conclusion, the integration of the Pseudo GPHS Labeling method with advanced ensemble models, especially CatBoost, proved to be the most effective strategy for DOY prediction. This approach delivered high accuracy and low error across stages and datasets, demonstrating its potential as a reliable tool for phenological forecasting in viticulture.

4.4. Model interpretation and visualization techniques

The SHAP violin plots (Fig. 2) for the XGBoost models applied to both the IEP and MR datasets highlight the relative importance of different features under supervised (SU) and semi-supervised (SM) learning approaches. Across all plots, GDD consistently emerges as the most influential feature, exhibiting a broad spread of SHAP values on both positive and negative ends. This indicates that GDD significantly impacts model predictions, both in increasing and decreasing the output, depending on its value. In contrast, features like Tmax, Tmin, and Tmed display more moderate impacts with their SHAP values clustering around zero, suggesting a smaller but still present influence on the model's output. Other features, including D30, CDR, and D35, show minimal variation in SHAP values across all plots, indicating they play a

Table 9
Performance of the models considered for the DOY. The reported metrics are R2 (higher is better) and RMSE (lower is better). The best result for each Dataset is bolded.

Stage	Datasets	Cat Boost		GBDT		RF		XGB	
		R^2	RMSE	R^2	RMSE	R^2	RMSE	R^2	RMSE
Flowering	CJ	**0.67 ± 0.32**	**0.72 ± 0.26**	0.44 ± 0.84	0.79 ± 0.40	0.35 ± 0.70	2.39 ± 0.51	0.28 ± 1.41	1.26 ± 0.53
	IEP	**0.94 ± 0.01**	**6.41 ± 0.30**	**0.94 ± 0.01**	**6.43 ± 0.47**	0.94 ± 0.01	6.54 ± 0.42	0.93 ± 0.01	6.60 ± 0.41
	PV	**0.95 ± 0.06**	**12.21 ± 0.22**	0.93 ± 0.14	14.70 ± 0.30	0.91 ± 0.05	18.15 ± 3.21	0.92 ± 0.03	15.80 ± 1.72
	MR	0.85 ± 0.03	7.57 ± 1.36	**0.86 ± 0.05**	**7.24 ± 0.76**	0.73 ± 0.10	10.87 ± 0.83	10.87 ± 4.83	10.87 ± 4.83
	IES	**0.93 ± 0.05**	**6.94 ± 3.0**	0.92 ± 0.04	7.23 ± 2.72	0.92 ± 0.06	7.15 ± 3.27	**0.93 ± 0.05**	**7.11 ± 2.94**
	SM	**0.63 ± 0.08**	**6.16 ± 0.62**	0.57 ± 0.08	6.65 ± 0.54	0.61 ± 0.12	6.27 ± 0.54	0.56 ± 0.13	6.70 ± 1.04
	Merge	0.92 ± 0.06	5.98 ± 1.38	**0.93 ± 0.05**	**5.88 ± 1.90**	**0.93 ± 0.04**	**6.13 ± 0.98**	0.83 ± 0.22	7.95 ± 4.91
Veraison	CJ	**0.73 ± 0.10**	**16.04 ± 2.01**	0.71 ± 0.13	16.27 ± 1.88	0.70 ± 0.06	17.44 ± 3.91	0.68 ± 0.15	17.20 ± 2.90
	IEP	**0.94 ± 0.01**	**6.41 ± 0.30**	**0.94 ± 0.01**	**6.43 ± 0.47**	0.94 ± 0.01	6.54 ± 0.42	0.93 ± 0.01	6.60 ± 0.41
	IES	**0.73 ± 0.07**	**6.21 ± 0.86**	0.70 ± 0.09	6.51 ± 0.97	0.71 ± 0.07	6.40 ± 0.72	0.70 ± 0.08	6.54 ± 0.76
	MR	0.77 ± 0.05	6.14 ± 0.52	0.75 ± 0.07	6.36 ± 0.73	0.78 ± 0.05	6.10 ± 0.48	**0.86 ± 0.05**	**7.24 ± 0.76**
	PV	0.98 ± 0.02	5.08 ± 3.10	**1.00 ± 0.00**	**2.13 ± 0.95**	0.99 ± 0.01	4.19 ± 1.24	0.99 ± 0.01	2.98 ± 1.88
	SM	**0.68 ± 0.12**	**7.38 ± 1.98**	0.67 ± 0.09	7.43 ± 1.44	0.65 ± 0.10	7.73 ± 1.67	**0.68 ± 0.10**	**7.35 ± 1.31**
	Merge	0.97 ± 0.01	7.38 ± 1.45	**0.98 ± 0.01**	**6.08 ± 1.46**	0.95 ± 0.03	10.54 ± 3.80	0.97 ± 0.02	7.13 ± 3.30
Inflorescence emerge	CJ	**0.73 ± 0.10**	**16.04 ± 2.01**	0.71 ± 0.13	16.27 ± 1.88	0.70 ± 0.06	17.44 ± 3.91	0.68 ± 0.15	17.20 ± 2.90
	IEP	0.71 ± 0.19	5.52 ± 0.89	**0.75 ± 0.17**	**5.13 ± 0.86**	0.66 ± 0.19	6.09 ± 0.80	0.69 ± 0.19	5.71 ± 1.04
	IES	0.40 ± 0.16	8.20 ± 1.32	0.34 ± 0.10	8.64 ± 1.01	0.35 ± 0.15	8.53 ± 1.26	0.30 ± 0.23	8.80 ± 1.67
	MR	**0.59 ± 0.07**	**7.22 ± 0.83**	0.56 ± 0.10	7.44 ± 1.01	0.55 ± 0.11	7.45 ± 0.76	0.50 ± 0.11	9.49 ± 3.29
	PV	0.94 ± 0.04	14.25 ± 2.22	**0.94 ± 0.04**	**13.72 ± 2.30**	0.90 ± 0.05	19.14 ± 5.21	0.93 ± 0.04	14.81 ± 2.82
	SM	**0.63 ± 0.08**	**6.16 ± 0.62**	0.57 ± 0.08	6.65 ± 0.54	0.61 ± 0.12	6.27 ± 0.54	0.56 ± 0.13	6.70 ± 1.04
	Merge	0.97 ± 0.01	8.36 ± 0.88	**0.97 ± 0.01**	**7.55 ± 0.71**	0.96 ± 0.01	9.75 ± 1.40	0.97 ± 0.01	7.69 ± 0.68
Development of fruits	CJ	0.99 ± 0.00	4.53 ± 1.08	**0.99 ± 0.01**	**2.34 ± 0.37**	0.98 ± 0.02	3.85 ± 1.94	0.98 ± 0.02	2.69 ± 0.31
	IEP	**0.79 ± 0.04**	**5.82 ± 0.50**	0.77 ± 0.04	6.17 ± 0.45	0.76 ± 0.04	6.30 ± 0.47	0.77 ± 0.04	6.11 ± 0.32
	IES	**0.71 ± 0.17**	9.55 ± 7.60	0.69 ± 0.17	9.86 ± 7.50	0.65 ± 0.18	10.42 ± 7.78	0.51 ± 0.85	14.06 ± 7.90
	MR	**0.57 ± 0.19**	**13.73 ± 7.93**	0.57 ± 0.19	13.75 ± 7.88	0.54 ± 0.20	14.19 ± 8.14	0.39 ± 0.48	15.01 ± 7.40
	PV	0.99 ± 0.00	11.81 ± 1.80	**0.99 ± 0.00**	**10.49 ± 2.18**	0.99 ± 0.00	11.23 ± 2.39	0.99 ± 0.01	11.58 ± 3.58
	SM	0.73 ± 0.21	9.05 ± 8.07	0.71 ± 0.19	9.22 ± 7.66	0.67 ± 0.19	9.86 ± 7.73	**0.79 ± 0.08**	**5.52 ± 0.97**
	Merge	0.99 ± 0.00	7.17 ± 0.79	0.99 ± 0.00	7.01 ± 0.71	**0.99 ± 0.00**	**6.91 ± 0.63**	0.99 ± 0.00	7.15 ± 0.97

relatively minor role in model predictions regardless of the learning approach or dataset. This analysis suggests that GDD is a key predictor in the XGBoost model, while the other features contribute to a lesser extent, with only minor differences observed between the supervised and semi-supervised models in terms of feature importance distribution.

For further evaluation, we calculated GDD using two base thresholds, 10 °C (GDD10) and 5 °C (GDD5), and generated dependence plots (Fig. 3) to compare their respective contributions and effects on the model's estimations. The dependence plots highlight the distinct contributions of GDD5 and GDD10 to the model's predictions. For GDD5, SHAP values transition from negative to positive at higher values

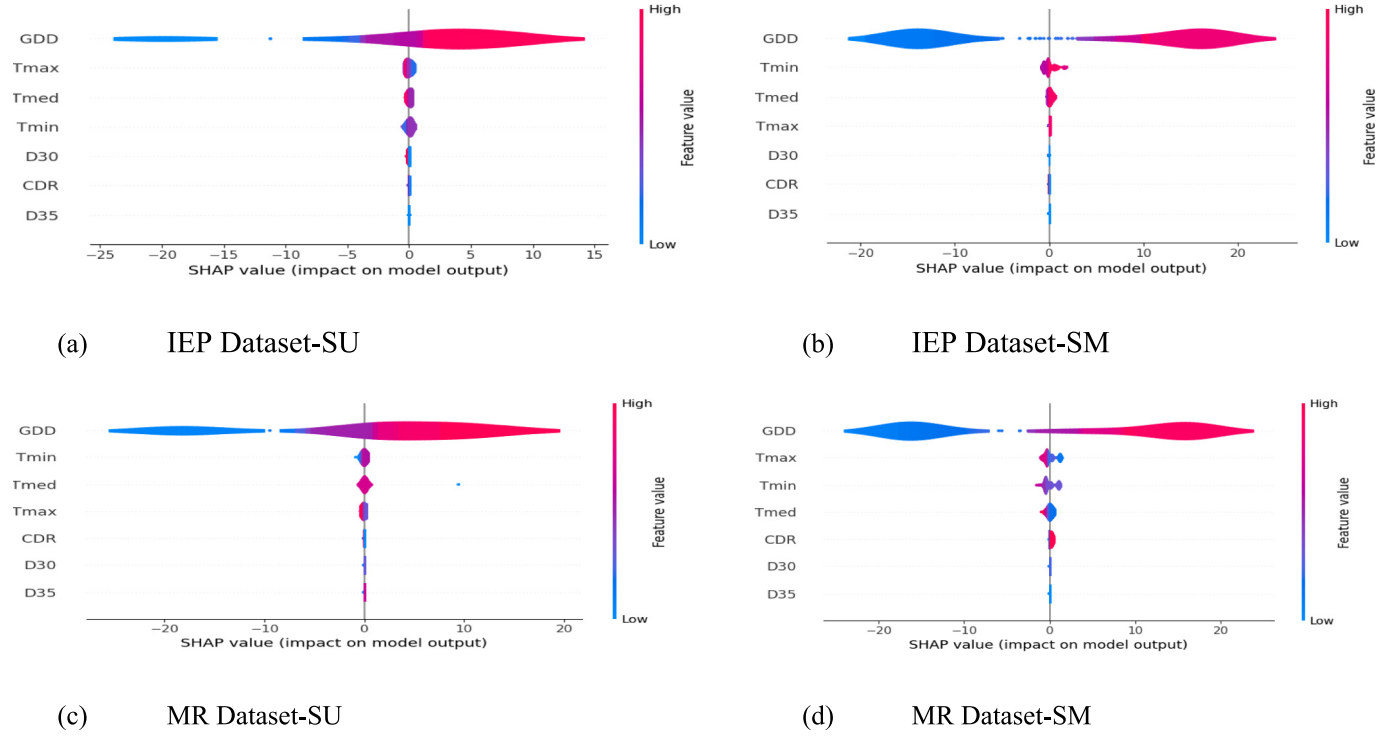
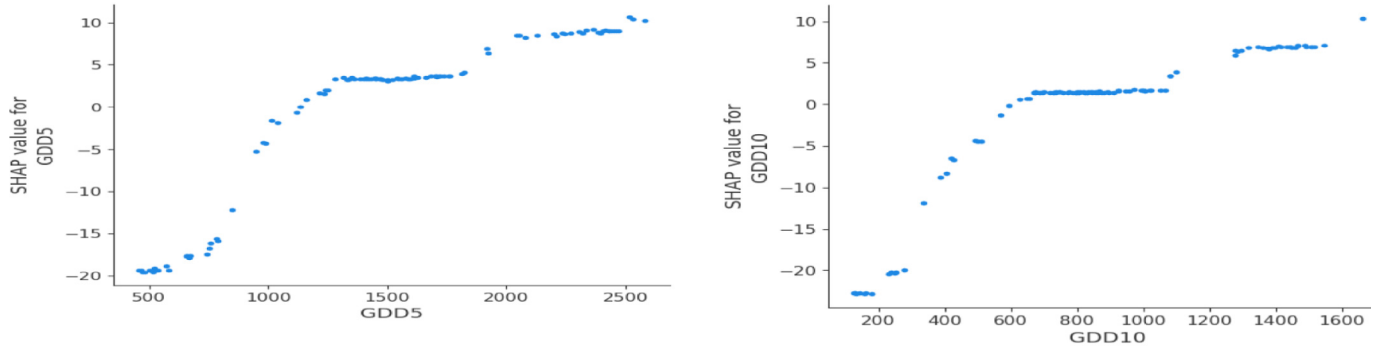


Fig. 2. SHAP Values in an XGBoost Violin Plot: Comparing SU (Supervised Learning) and SM (Semi-Supervised Learning) Approaches.



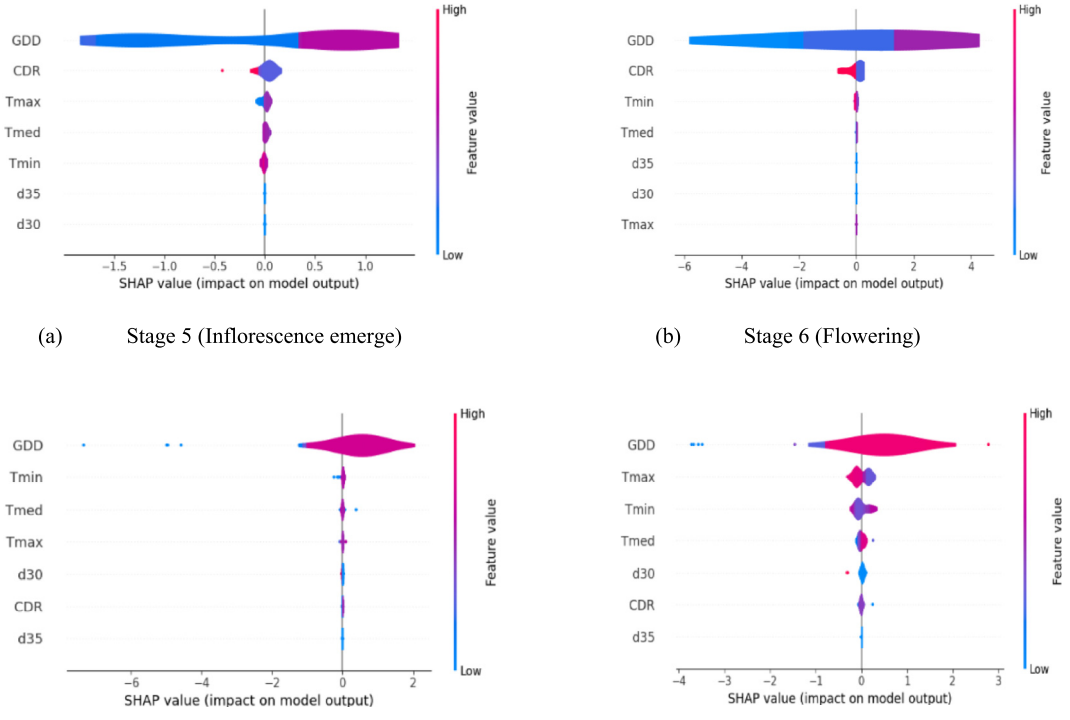
(a) GDD calculated with a 5°C threshold (b) GDD calculated with a 10°C threshold

Fig. 3. SHAP dependence plot showing the impact of different GDD thresholds (e.g., 5 °C and 10 °C) on model predictions for the IEP dataset.

(approximately 1000–1500), with a prolonged plateau between 1500 and 2000, followed by a steady rise beyond 2000. This suggests that GDD5 captures a wider range of temperature conditions and maintains consistent importance across broader climatic variations. In comparison, GDD10 exhibits a steeper transition from negative to positive SHAP values at lower ranges (around 400–800), with a plateau occurring earlier (800–1000), and a more localized rise at higher values, indicating its relevance in warmer climates or specific growth stages. These patterns underscore GDD5's broader influence on model predictions and GDD10's sharper, more focused impact.

Furthermore, Fig. 4 presents the distribution of SHAP values across BBCH grape growth stages (5, 6, 7, and 8), offering a deeper evaluation of how feature influence shifts throughout the grapevine's developmental phases. Across all growth stages, GDD remains consistently important, highlighting the role of cumulative temperature throughout grape development. In Stage 5 (Inflorescence emergence), CDR is highly impactful, second only to GDD, while d30 and d35 show minimal influence. In Stage 6 (Flowering), d30 and d35 gain slightly more importance than the previous stage, and CDR has a reduced effect. By Stage 7 (Development of fruits), specific temperature metrics like Tmin and Tmed become more prominent, while CDR, d30, and d35 lose significance. In Stage 8 (Ripening of berries), GDD remains impactful, with a balanced influence from Tmax, Tmin, and Tmed, indicating a reduced effect from specific temperature metrics. Overall, GDD consistently drives growth, while CDR and temperature deviations (d30 and d35) have stage-specific importance, peaking in the early stages and during flowering.

In our study, we evaluated the predictive accuracy of both supervised (SU) and semi-supervised (SM) learning approaches using the XGBoost model across three datasets. The regression plots in Fig. 5



(a) Stage 5 (Inflorescence emerge) (b) Stage 6 (Flowering)

(c) Stage 7 (Development of fruits) (d) Stage 8 (Ripening of berries)

Fig. 4. SHAP Values in an XGBoost Violin Plot: This figure shows the distribution of SHAP values across BBCH grape growth stages (5, 6, 7, and 8), highlighting how feature influence varies at each developmental stage in the XGBoost model for the IEP dataset.

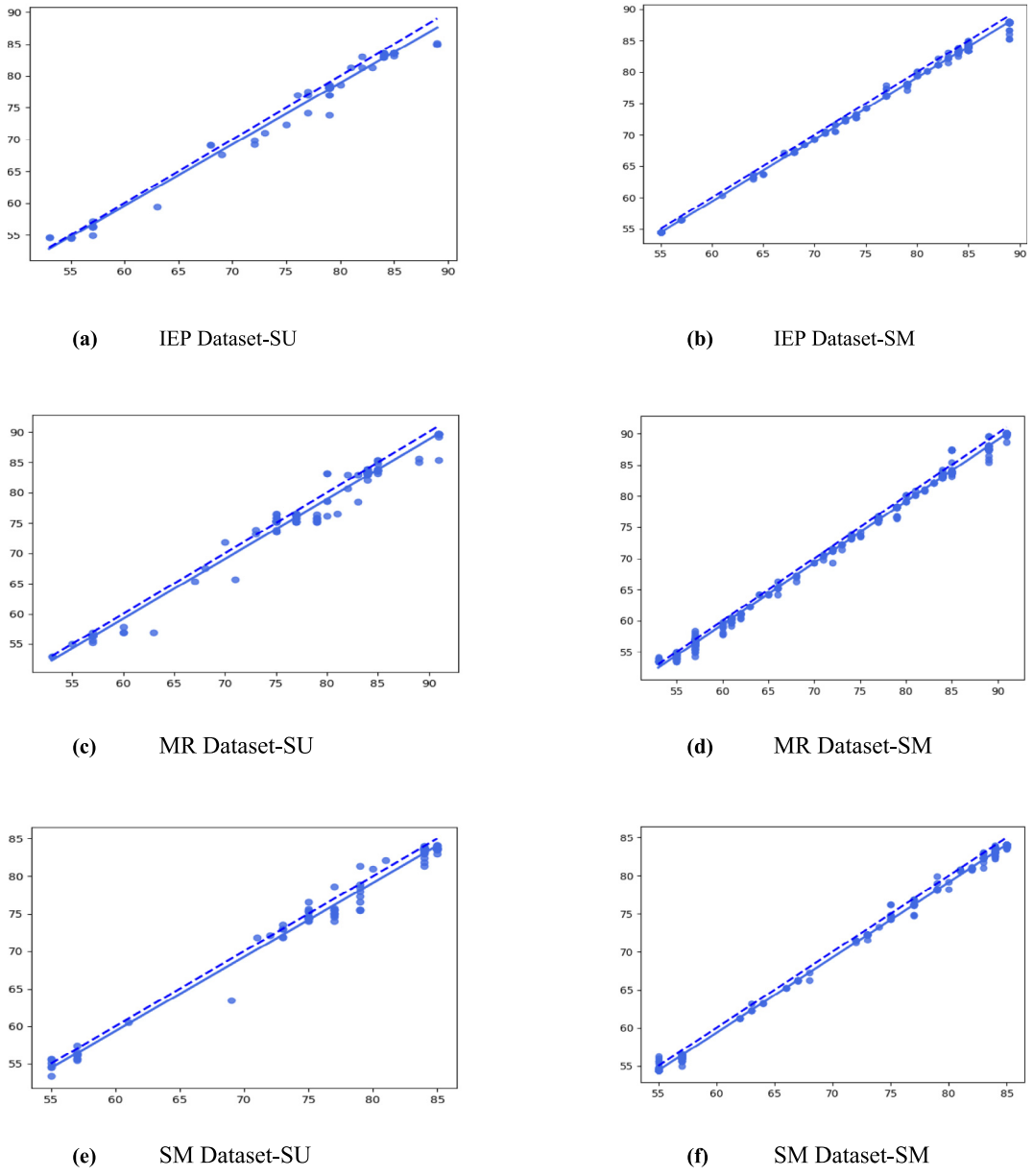


Fig. 5. Real vs. Predicted BBCH Stages for XGBoost Model across Datasets: Premariacco (IEP, plots (a) and (b)), Manzano (MR, plots (c) and (d)), and Cividale del Friuli (SM, plots (e) and (f)). The x-axis shows Actual Labels, and the y-axis shows Predicted Labels. The dashed line represents the ideal 1:1 line where predictions perfectly match actual BBCH labels, while the solid line represents the regression line between predicted and actual labels, indicating the model's trend. Data points illustrate the alignment between actual and predicted labels, with SU representing supervised learning and SM representing semi-supervised learning approaches.

illustrate the relationship between actual and predicted BBCH stages, where the dashed line represents the ideal 1:1 alignment and the solid line represents the model's fitted regression line. The results indicate that the semi-supervised approach outperformed the supervised method across all datasets, with a closer alignment of data points to the 1:1 line in the SM plots. This tighter clustering along the ideal line in the semi-supervised approach underscores its higher accuracy in predicting BBCH stages. Key metrics such as R^2 and RMSE also showed significant improvement, affirming the advantage of incorporating unlabeled data into the model. Overall, these findings suggest that the semi-supervised model better captures the complexity of growth stage prediction, providing more reliable predictions across varied datasets. This improvement underscores the value of semi-supervised learning in agricultural growth modeling and highlights its potential for enhancing predictive accuracy in similar applications.

5. Discussion

We used Shap Values (refer to Fig. 2–4) as an explainable model to understand the importance of different features in our machine learning models. Shap Values help us clarify how each input feature affects the model's predictions, improving our understanding of how the model makes decisions. SHAP values measure the change in the model output when a variable's components are perturbated *(Lundberg et al., 2019)*. From Shap values, we can observe that GDD is by far the most important variable. While other variables may interact to augment the model's predictive capacity, the role of GDD in influencing phenological changes remains central.

Previous studies utilizing statistical models have thoroughly documented the significance of GDD in predicting GPHS. Zapata et al. *(Zapata et al., 2017)* predicted the phenological stages of budbreak,

bloom, and veraison in 17 different cultivars by parameterizing the base temperature of GDD. In a similar approach, phenological stages were evaluated in four cultivars under Mediterranean conditions (Ortega-Farias and Riveros-Burgos, 2019). In recent years, efforts have been made to enhance the cumulative prediction of degree days under specific conditions (Caffarra and Eccel, 2010). Traditional statistical models often lack the flexibility to adapt to specific local conditions and climate variations. To address this, we enhanced the GDD approach by integrating environmental measurements directly into a machine learning model. This approach allows testing and parameterization across diverse varieties and geographic areas, improving adaptability and robustness. While GDD captures accumulated heat, which is crucial for plant development, adding daily maximum (Tmax), minimum (Tmin), and median (Tmed) temperatures offers a nuanced view of temperature's influence on phenology (Hatfield and Prueger, 2015). Tmax and Tmin account for temperature extremes affecting growth and stress responses, while Tmed reflects daily variability, enhancing the model's accuracy in capturing temperature effects on phenological phases (Hatfield and Prueger, 2015). This integration gives the model greater sensitivity to temperature dynamics, boosting its predictive power across diverse climate conditions.

Our research highlights the advantages of the pseudo-labeling approach in predicting Day of Year (DOY) and Grapevine Phenological Stages (GPHS), achieving considerable improvements in accuracy and robustness over traditional models. The semi-supervised model improves R^2 and RMSE metrics, as shown in regression plots and comparative tables, effectively capturing complex climatic interactions for reliable GPHS predictions. This advancement is especially evident in DOY, where accurate timing plays a crucial role in optimizing vineyard operations such as disease management, irrigation, pruning, and harvesting, ultimately enhancing crop yield and quality (Mucalo et al., 2024).

Furthermore, the proposed approach demonstrated significant improvements in predictive accuracy for DOY predictions for flowering on the IEP dataset, particularly with the CatBoost model, which achieved an R^2 of 0.94 $\pm$ 0.01 and an RMSE of 6.41 $\pm$ 0.30. This performance surpasses that of previous models, such as Piña-Rey et al. (Piña-rey et al., 2021) employed a GDD-based approach with an R^2 of approximately 0.82 for flowering predictions. Also, Rafique et al. (Rafique et al., 2024) reported R^2 values ranging from 0.90 to 0.94 for DOY predictions using a temperature-based model, effectively capturing the timing of key phenological stages of table grapes under South Asia's unique climatic conditions. Schrader et al. (Schrader et al., 2020) developed multifactor models incorporating variables like cultivar, day of the year, soil thaw date, photoperiod, GDD, soil degree days, and solar accumulation, achieving R^2 values of 0.996 for budbreak, 0.985 for bloom, 0.985 for veraison, and 0.869 for harvest. However, these models were tailored for cold-climate wine grapes and may not be generalized across different climatic conditions. It is important to note that direct comparisons between models should be approached with caution, as variations in dataset composition, grape variety, and climatic conditions can make exact equivalences challenging to establish.

As Min et al. (Min et al., 2022) emphasize, semi-supervised learning allows for leveraging vast unlabeled data, enhancing model generalization across regions, which is particularly valuable for vineyards with limited labeled data. This scalable solution enhances DOY and GPHS predictions, supporting data-driven approaches and precision practices in viticulture. The robustness of our approach advances vineyard management practices by providing a non-invasive, cost-effective tool that enables strategic timing for flowering and ripening stages, underscoring the model's potential in complex viticulture environments.

6. Conclusion

The precise Prediction of grapevine phenological stages (GPHS) is essential, as it carries significant operational implications for crop management, yield estimation, and overall agricultural planning. Common state-of-the-art solutions for GPHS involve the usage of techniques based on image processing. However, this leads to intrusive solutions, requiring positioning multiple cameras in the vineyard, which are also difficult and costly to maintain. Moreover, the data collection process for GPHS is difficult and expensive, as detecting the precise phenological stage requires experts to assess it directly in the field. To address these two challenges, we introduced the Pseudo GPHS Labeling strategy, a semi-supervised approach designed to leverage readily available, non-intrusive climatic data. This method also tackles missing values in data annotations by automatically generating pseudo-labels. Our research evaluated the impact of the proposed strategy on the performance of eight machine learning algorithms, employing metrics such as coefficient of determination (R^2) and root-mean-square error (RMSE). This comprehensive evaluation encompassed seven distinct datasets, each with varying percentages of missing values. The results of our study are promising, with substantial performance improvements, highlighting the effectiveness of the semi-supervised approach in mitigating the impact of missing data. This approach shows great potential for advancing the precision and reliability of GPHS predictions. Moreover, our focus on climatic data-based predictions presents several advantages over image-based methods. This approach is cost-effective, non-intrusive, and adaptable to vineyards of different scales and technological capabilities, making it a practical choice for the viticulture industry. In essence, this research addresses a pressing need in the viticulture domain, where the accuracy of GPHS predictions significantly impacts vineyard management and wine production. Although the proposed method led to promising results, the main limitation of our approach lies in the absence of other environmental factors, such as soil moisture, solar radiation, and vineyard-specific management practices. However, our encouraging results suggest that these challenges can be addressed in future work by expanding the dataset to include a broader range of environmental variables and training the model to incorporate this heterogeneity, thereby enhancing its adaptability to diverse vineyard conditions.

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Author statement

We wish to confirm that there are no known conflicts of interest associated with this publication and we also confirm that the manuscript has been read and approved by all named authors. Besides, there are no other persons who satisfied the criteria for authorship but are not listed.

We further confirm that the order of authors listed in the manuscript has been approved by all of us and our role in the CRediT authorship contribution statement section in the manuscript is mentioned accurately.

In addition, we confirm that we have given due consideration to the protection of intellectual property associated with this work and that there are no impediments to publication, including the timing of publication, concerning intellectual property. In so doing we confirm that we have followed the regulations of our institutions concerning intellectual property.

We understand that the Corresponding Author is the sole contact for the Editorial process (including the Editorial Manager and direct communications with the office). He is responsible for communicating with the other authors about progress, submissions of revisions, and final approval of proofs.

CRediT authorship contribution statement

Mehdi Fasihi: Writing – review & editing, Writing – original draft, Software, Methodology. **Mirko Sodini:** Writing – review & editing, Supervision. **Alex Falcon:** Writing – review & editing. **Francesco Degano:** Data curation. **Paolo Sivilotti:** Writing – review & editing, Visualization, Supervision, Project administration. **Giuseppe Serra:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Description of eight supervised learning algorithms

A.1. Multiple linear regression (MLR)

MLR is a regression model that calculates the relationship between a dependent variable and two or more independent variables using a straight line. It applies various explanatory variables to estimate the outcome of a quantitative dependent variable (Uyanık and Güler, 2013). In Eq. (3), y is the target variable (GPHS); $x_0, x_1, \ldots\ldots, x_n$ are the independent input features (climatic data information), $\beta_0, \beta_1, \ldots\ldots, \beta_n$ are the fitting parameters, and n in this research is 7 (the number of attributes).

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \ldots + \beta_n x_n \tag{3}$$

A.2. Random forest (RF)

RF algorithm is a robust and influential tool in machine learning known for its capability to generate multiple decision trees during training (Ho, 1998). Its effectiveness stems from its ability to efficiently handle many input variables, enabling it to analyze complex data patterns accurately. Eq. 4 shows the formula for RF in which y is the target variable, $X\{x_0, x_1, \ldots\ldots, x_n\}$ are climatic data information, and K refers to the number of decision trees (determined by hyperparameter tuning), and $Tree_k(X)$ represents the prediction of $k - th$ decision tree.

$$y = \frac{1}{K}\sum_{k=1}^{k} Tree_k(X) \tag{4}$$

A.3. Gradient boosting decision tree (GBDT)

GBDT is a prominent machine learning algorithm based on the Gradient Boosting methodology, an iterative process grown in a greedy, forward stage-wise manner, focusing on reducing the errors made by previous models in the ensemble. The algorithm follows a well-defined sequence. Initially, the ensemble model's prediction is set to the average of the target values. Then, at each iteration, it fits a weak learner, i.e. a simple and relatively low-complexity model to predict the "error" of the current ensemble model and adjust it using the negative gradient direction (Friedman, 2001). The newly trained weak learner is then added to the ensemble, making it more robust. In the context of GBDT, the weak learner is a decision tree. The updated formula at each iteration m is given by Eq. 5:

$$y_m(X) = y_{m-1}(X) + \gamma.h_m(X) \tag{5}$$

Here, $y_m(X)$ is the GPHS prediction at iteration m, $y_{m-1}(X)$ is the prediction from the model up to the $m - 1th$ iteration, $h_m(X)$ is the decision tree fitted to the error at the $m - th$ iteration, and γ is the learning rate (Huang et al., 2022). The final GPHS prediction is achieved by summing up the prediction of all the weak learners according to Eq. 6:

$$y(X) = y_0(X) + \gamma.h_1(X) + \gamma.h_2(X) + \ldots + \gamma.h_M(X) \tag{6}$$

A.4. eXtreme gradient boosting (XGBoost)

XGBoost is an extension of GBDT and employs a distributed array of multiple decision trees to address classification or regression challenges (Chen and Guestrin, 2016). One notable difference of this algorithm from conventional GBDT is the incorporation of a regularization component within the objective function. This component enforces a constraint on generalization performance, effectively curbing overfitting and thereby enhancing the model's capacity to extrapolate to novel data instances (Huang et al., 2022). Eq. 7 shows the general learning objective for this model in which n is the number of training instances, y_i is the true label for the i-th instance and $\tilde{y}$ is the predicted output for the ith instance. Besides, in this Equation k is the number of trees in the model, f_k is the k-th tree and ∂f_k is the regularization term that penalizes complex trees to prevent overfitting.

$$ObjectiveFunction = \sum_{i=1}^{n}\left(y_i - \tilde{y}_i\right)^2 + \sum_{k=1}^{k}\partial(f_k) \tag{7}$$

A.5. Categorical Boosting (CatBoost)

CatBoost emerges as a prominent contender within the realm of boosting algorithms, presenting itself as an alternative to XGBoost. An intrinsic attribute of CatBoost is its proficiency in managing categorical features through a permutation-based strategy, fostering the potential for enhanced accuracy compared to conventional algorithms (Prokhorenkova et al., 2017). Beyond this, CatBoost introduces a range of further enhancements over preceding boosting models, encompassing streamlined hyperparameter tuning and expedited processing when compared to XGBoost (Huang et al., 2022). Mathematically, the GPHS prediction obtained with CatBoost is similar to Eq. 4; hence, for simplicity, it is not shown here.

A.6. Multilayer perceptron (MLP)

MLP model was chosen from the realm of artificial neural networks to facilitate the implementation of a diverse array of models, comprising input, output, and one hidden layer. The MLP operates as a feedforward neural network, aiming to generate a value close to the target during the training phase. It then fine-tunes a weight matrix to establish a correlation between the input and output variables (Subasi, 2020). Eq. 8 shows the general formula for the GPHS prediction of a neural network in which σ_1 and σ_2 are non-linear activation functions, W_{I-H} and W_{H-O} refer to the learnable weight matrices standing between the input-hidden and the hidden-output layers, and B and b are learnable bias terms.

$$y = \sigma_2(W_{H-O}\sigma_1(W_{I-H}\cdot X + B) + b) \tag{8}$$

A.7. Support vector regression (SVR)

Support Vector Regression (SVR) is grounded in the Vapnik–Chervonenkis (VC) dimension theory, aiming to balance model complexity with learning capacity for optimal performance (Cortes et al., 1995). Unlike traditional regression techniques, SVR seeks to find a function that fits within a margin of error rather than exactly through

all data points. In its essence, SVR identifies a decision function (or hyperplane in higher-dimensional spaces) that optimally aligns with input climatic data, allowing for some tolerance of error, controlled by a regularization parameter. This is accomplished by transforming input data into a higher-dimensional feature space, enabling linear relationships to approximate non-linear patterns effectively. To achieve this, a Radial Basis Function (RBF) kernel is employed to model and capture non-linear relationships within the data efficiently. Eq. (9) shows the SVR model output in which a_i and a_i^* are from the optimization problem, b is a constant offset to ensure the predicted values align with the data, and $K(x_i, x)$ is the kernel function (e.g., RBF, polynomial).

$$f(x) = \sum_{i=1}^{n} (a_i - a_i^*) K(x_i, x) + b \tag{9}$$

A.8. K-nearest neighbors (KNN)

KNN is a non-parametric method that determines its output by averaging the values of the k "closest neighbors" to a given input object (Cover and Hart, 1967). These neighbors are selected based on their similarity to the input object in terms of features. Parameter k is crucial in obtaining an accurate estimation of the target variable. The distance between the new data point and all other data points (Eq. (10)) in the training set is calculated based on c, where n is the number of features, and x_i and y_i are the values of $i - th$ feature for the new data point and a point in the training set, respectively.

$$Distance = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2} \tag{10}$$

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